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Local control and visual acuity following treatment of medium-sized ocular melanoma using a contact eye plaque: A single surgeon experience

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ABSTRACT

PURPOSE: To evaluate the outcomes of choroidal melanoma in (CM) patients treated with ^{125}I episcleral plaque brachytherapy and to compare our single surgeon results with the multi-institutional Collaborative Ocular Melanoma Study (COMS).

METHODS AND MATERIALS: A review was performed of all CM patients treated with ^{125}I episcleral plaque brachytherapy by ophthalmologist in accordance with established COMS guidelines.

RESULTS: The records of 35 patients were reviewed. The median longest basal tumor diameter and apical tumor height was 13.5 and 7.8 mm, respectively. Median dose to the apex was 8609 cGy at a median dose rate of 92 cGy/h. At a median followup of 45 months, 35 patients had local control and 33 had successful organ preservation. At 5 years, the local control rate was 100%, and the eye preservation rate was 94%. Five patients developed hepatic metastasis at a median of 58 months, and 2 succumbed from disease. The 5-year survival rate was 84%, and the 5-year rate of death with histopathologically confirmed metastasis was 15%. Of the 22 patients with at least 3 years of followup, 68% had a visual acuity in the treated eye of 20/200 or worse.

CONCLUSION: Excellent local control, eye preservation rates, and survival outcomes following ^{125}I episcleral plaque application for CM can be optimized by having an experienced ophthalmologist place the plaques. Additionally, hepatic metastasis can occur more than 5 years postimplant regardless of local control; therefore, longer systemic staging should be considered.

Keywords: Ocular; Melanoma; COMS; Plaque; Eye; Choroidal; Brachytherapy

Introduction

Ocular/choroidal melanoma (CM) is a rare and potentially fatal disease, requiring significant treatment expertise at a specialty center. Although the incidence is only 4.3 cases/million population (1), it is the most common primary malignancy of the globe and failure to locally control the disease leads to vision loss, systemic metastasis, and death. Until the 1980s, the standard of care was surgical enucleation for all CM, despite its disfiguring cosmesis, effect on quality of life, and obvious vision loss associated with enucleation (2). Alternative organ sparing modalities, such as radiation therapy (i.e., episcleral brachytherapy or charged particle external beam radiotherapy), transpupillary thermotherapy, photocoagulation, and cryotherapy have since been developed to preserve the affected eye and possibly vision without sacrificing the survival benefits associated with enucleation (3, 4).

The largest randomized globe preservation trial to date challenging enucleation as the standard of care for CM was the Collaborative Ocular Melanoma Study (COMS). In this multi-institutional trial, more than 1300 patients with medium-sized choroidal tumors were randomized to either enucleation or ^{125}I episcleral plaque (5). No survival differences were observed between the two arms after 12 years of followup, whether considering death from all causes or

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death with histopathologically confirmed melanoma metastases (6). Eighty-seven percent of the patients treated with ^{125}I brachytherapy retained their eye for 5 years with 43% of irradiated eyes having a visual acuity of 20/200 or worse at 3 years (7, 8). Other authors have reported similar local control rates and visual acuity changes (9, 10).

Since 1996, our institution has had a single ophthalmologist apply the ^{125}I episcleral plaque based on COMS design for treatment of ocular melanoma. Here, we review this experience and compare our single surgeon results with those previously published in the literature.

Methods and materials

Between August 1996 and October 2008, 35 patients with CM underwent ^{125}I episcleral plaque brachytherapy. Institutional review board approval was obtained before the study. Of these patients, all were Caucasian and 17 were male and 18 female. The median age was 68 years (range, 43−85 years.) and the initial visual acuity in the affected eye was 20/200 or worse in 10 patients (28%) (Table 1).

The median longest basal tumor diameter was 13.5 mm (range, 7−18 mm) and apical tumor height was 7.8 mm (range, 3−13.5 mm). Four tumors were noted to have ciliary body involvement, although none were predominantly ciliary body in origin. No tumors were contiguous with the optic disk and all had less than 2 mm of extrascleral extension.

All plaques were of COMS design with a median size of 18 mm (range, 14−20 mm). The median prescription dose to the apex using these plaque implants was 8609 cGy (range, 7500−10,500 cGy) delivered at a median dose rate of 92 cGy/h (range, 77−140 cGy). The median implant duration was 96 h (range, 72−105 h).

The diagnosis of CM was made or verified by a single ophthalmologist on the basis of a comprehensive ophthalmologic evaluation, ultrasonography, and/or fluorescein angiography. Patient characteristics (age, sex, race, preexisting ophthalmic conditions, and preimplant best corrected Snellen chart visual acuity) and tumor characteristics (longest basal diameter, maximum apical height, and ciliary body involvement) were recorded. A pretreatment metastatic workup, including chest X-ray and liver function tests (if indicated, CT of the chest, abdomen, and pelvis), was negative in all patients.

All plaques were of COMS specifications (11) and chosen to encompass a minimum 2 mm margin around the tumor with the exception of the single 18 mm lesion. Only a 1 mm margin could be achieved because of plaque size limitations. Using a predesigned seed carrier, the corresponding number and activity of ^{125}I seeds were then chosen to deliver 85−100 Gy precisely to the apex at a projected dose rate of 85 cGy/h (range, 50−125 cGy/h). The ophthalmologist then applied the plaque as previously published by COMS (11). Plaque diameter, dose to the apex, dose rate, and implant duration were recorded.

Followup examinations were performed at 1, 3, and then every 6−12 months thereafter. Indirect ophthalmoscopy, ultrasonography, and occasional fluorescein angiography were performed at the discretion of the ophthalmologist to assess tumor response. Tumor progression was defined as any enlargement of the tumor and treated with enucleation. Best corrected visual acuity in the affected eye via Snellen chart, treatment related toxicity (i.e., retinopathy, optic neuropathy, and retinal detachment), and development of metastases were also recorded.

The 5-year actuarial local control, eye preservation rate, survival, and death with histopathologically confirmed metastasis were calculated using the Kaplan−Meier method (12).

Table 1
Initial patient characteristics

Patient characteristics	n
Gender (n = 35)	
Male	17
Female	18
Age (y)	
Median	68
Range	43−85
Race	
Caucasian	35
Visual acuity (affected eye)	
<20/200	25
≥20/200	10

Results

At a median followup of 45 months (range, 1−159 months), 35 patients (100%) had local control, 33 (94%) had achieved successful organ preservation, and 32 (91%) were alive. The 5-year actuarial local control rate was 100% and 5-year actuarial eye preservation rate was 94%. Two patients required enucleation at 24 and 30 months postimplant for pain and neovascular glaucoma, respectively. Five patients developed hepatic metastasis at median of 58 months (range, 19−108 months), and two died of disease within 6 and 35 months, respectively. The remaining 3 patients are alive with disease as of this submission. The 5-year actuarial survival rate was 84% and the 5-year rate of death with histopathologically confirmed metastasis was 15%. One patient developed two metachronous ipsilateral CM distant to the original lesion, which was effectively controlled by repeat ^{125}I episcleral plaque application. No patients developed other malignancies.

Including the eyes that were enucleated, the visual acuity was 20/200 or worse in 77% of the patients at a median of 45 months. Among the patients with 3 years of followup, 68% had a visual acuity of 20/200 or worse in the treated eye (Table 2).

Table 2
Post-treatment visual acuity

Time postimplant	Patient record available	<20/200	≥20/200 *n* (%)
1 y	32	13	19 (59)
2 y	27	11	16 (60)
3 y	22	7	15 (68)
45 mo (median)	35	8	27 (77)

Retinal detachment was documented postprocedure in 12 patients (34%) and 11 patients (31%) were identified to have radiation retinopathy. Three patients (9%) developed radiation maculopathy and optic neuropathy occurred in 1 patient (3%). Eight patients (23%) developed cataracts in the treated eye, which developed following the visual acuity change in all patients.

Discussion

The largest globe preservation trial to date, the COMS, used ^{125}I episcleral plaque brachytherapy. This trial randomized more than 1300 patients with medium-sized CMs at 1 of 43 North American COMS centers to an ^{125}I plaque or enucleation. At 12 years followup, no statistically significant differences between the two arms were observed in overall survival (43% vs. 41%, respectively) or death with histopathologically confirmed melanoma metastasis (21% and 17%, respectively) (6). Eighty-seven percent of the patients treated with ^{125}I brachytherapy retained the eye for 5 years, and 57% maintained visual acuity of better than 20/200 at 3 years (7, 8). The risk of treatment failure at 5 years was 10.3% and was the most common reason for enucleation within 3 years (7). Although 1 of our patients developed metachronous CMs, they were distant to the primary and each other and were not felt to represent marginal misses.

In our study, no treatment failure has been observed and we achieved a 94% organ preservation rate at a median followup of 45 months. These numbers compare favorably with the COMS data and are more in line with the 94% eye preservation rate and 93% 5-year local control achieved by Jensen *et al.* (13) in their retrospective review of 156 patients treated with COMS designed ^{125}I episcleral plaques at the Mayo Clinic. This group used only two ophthalmologists, which may explain why the slightly higher local control and eye preservation rates than those in the COMS study, in which patients were referred to ophthalmologists at 1 of the 43 COMS clinical centers in the United States and Canada (5). Methods to ensure proper plaque placement are critical to successful radiation therapy (14–18), and having an experienced ophthalmologist on staff not only allows for uniformity of diagnosis but also of plaque application as well. The retrospective nature of both our study and that of the Mayo group is hampered by potential inherent bias and uncontrolled followup data collection variables. However, our 5-year actuarial overall survival of 84 % and metastasis-free survival of 85% data are consistent with the Mayo groups' respective 5-year estimated rates of 82% and 91% and COMS 5-year respective rates of 81% and 89% (6, 13).

Despite the benefit of organ preservation associated with ^{125}I episcleral plaque brachytherapy, preservation of useful visual function, 20/200 or better, is not always achievable. Sixty-eight percent of our patients with documented visual acuity at 3 years had a visual acuity 20/200 or worse in the treated eye. This is an outcome variable that the COMS authors reported and 43% of their treated eyes at 3 years had a visual acuity 20/200 or worse (8). A factor that may account for the superior outcome of the study group is the fact that only 10% of their patients had a visual acuity of 20/200 or worse in the affected eye at presentation, compared with 28% in our series (8). Also median apical height of our tumors was larger compared with the COMS group (7.8 mm vs. 4.8 mm, respectively) (5). Larger apical height has been reported in COMS report 16 to be strongly associated with posttreatment visual acuity of 20/200 or worse (8). The Mayo group study had a similar visual acuity outcome to ours, documenting that 44% of their patients had a visual acuity of 20/200 or better in the treated eye, despite 92% having 20/200 or better at presentation (13). By comparing 20/200 or better in the treated eye, instead of 20/200 or worse, 60% of our patients with documented visual acuity at 3 years maintained this level of acuity. This result is similar to the 58% value achieved by Quivey *et al.* (19) in 239 patients treated with ^{125}I plaques. Correcting for the above, all visual acuity data we report are comparable with other published literature.

The causes of visual loss after plaque application are likely multifactorial, including tumor recurrence, cataract formation, and excessive dose to critical ocular structures, including the macula or optic nerve (20). Brady and Hernandez (21) reported 90% of their patients receiving <50 Gy to the fovea maintained a post-treatment visual acuity of 20/200 or better vs. only 52% of patients with fovea doses >50 Gy. The COMS report suggested greater baseline tumor apical height and shorter distance between the tumor and the foveal avascular zone as factors that were strongly associated with visual acuity of 20/200 or worse (8). Jones *et al.* (20) also showed that higher dose rates to the macula correlated strongly with poorer post-treatment visual outcome, and dose rates of greater than 111 cGy/h were associated with a 50% risk of significant vision loss (>2 lines on a Snellen eye chart). Unfortunately, multiple centers have shown that lower dose rates to the tumor, which may in theory spare visual decline, have been linked to higher rates of local failure, metastasis, and mortality (13, 22, 23). Gunduz *et al.* (23) and Quivey *et al.* (22) specifically showed that reduced tumor apex dose rates of <75 cGy/h correlated with increased risk of systemic spread of disease and local failure, respectively.

A surprising outcome of our study was that median time to the development of hepatic metastasis in our patients was 58 months (range, 19–108 months). Interestingly, 3 of 5 of these patients developed hepatic metastasis more than 5 years after their implant. This raises questions of how long and how best to follow patients with otherwise locally controlled CM. COMS reported that by 5 years after enrollment, 189 (14%) of the 1317 patients had been diagnosed with melanomatous metastasis, and by 10 years 182 (23%) of the 799 surviving patients had developed metastases (6). Perhaps CM patients should have annual restaging enhanced CT scans of the liver, especially given the new era of oligometastatic disease control with local modalities, that is, stereotactic body radiotherapy, radioactive TheraSpheres (Theragenics Corp., Buford, GA), radioablation, and surgery. No cure yet exists for systemic spread, and death typically ensues in months (13).

Conclusion

Excellent local control, eye preservation rates, and survival outcomes following ^{125}I episcleral plaque application for CM can be optimized under the service of an experienced ophthalmologist. Unfortunately, organ preservation does not guarantee preservation of function and this feature requires careful consultation with the patient. Additionally, hepatic metastasis can occur more than 5 years postimplant regardless of local control, and longer and more thorough systemic staging should be entertained.

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